

CS/EE120B – Custom Laboratory Project

Report

Project Title: Flappy Bird

Name: Peter Hsia (phsia006)

Date: December 2025

Introduction

The project I did was an endless game of Flappy Bird on the Arduino uno. The goal of the game was to see how high of a score you can get as the bird needs to dodge the pipes by going through them while being able to control him flying upwards, the challenge of the game is the bird due to gravity will want to fly downwards so its your job to make sure that does not happen. The game ends when the bird touches a pipe, the ground, or the ceiling.

We have a 16×2 LCD displaying the title screen, score, and game-over messages.

The 1.44" SPI TFT LCD handles the animated graphics, including sky color, custom sprite(bird), and pipe movement. My design has smooth animation of the bird and pipe movement with custom multi color sprites.

A **joystick module** is used for player control, allowing the user to flap upward while gravity continuously pulls the bird downward.

All major objectives were met. The game runs smoothly and all game logic executes entirely on the AVR without external controllers.

Build-Upons

Build-Upons Implemented:

1. Use of a 16x2 display + Custom character

Successfully used as the game uses the 16x2 as a msg to start the game with a **smiley face custom character**, keeps score and an game over screen

2. ST7735 TFT LCD

Implemented a driver (color, fill screen, set window, sprite drawing) using custom SPI from scratch.

3. Custom sprite on TFT LCD

Draw own custom bird then converted PNG sprites into raw RGB data and wrote a 2× scaling renderer to display 24×18 pixel Flappy Bird at 48×36 in-game resolution.

User Guide

Controls:

Joystick Button(Press in) (A1, SW pin)

From title screen -> starts the game.

After Game Over -> restarts the game.

Joystick Push up Button (A3)

Push upward -> bird flaps upward(holding up makes it keep going up).

Release -> gravity pulls bird downward(slower).

Game logic

The bird flies through pipes trying to dodge them by going through, if you hit the pipes at all or touch the ground/ceiling you automatically lose

Each time you successfully go through a pipe, you get +1 to your score. To start the game you press in the joystick and when you lose, you can restart by pressing the button again

The Score will be displayed on the 16x2 while playing and when game ends

Hardware Components Used

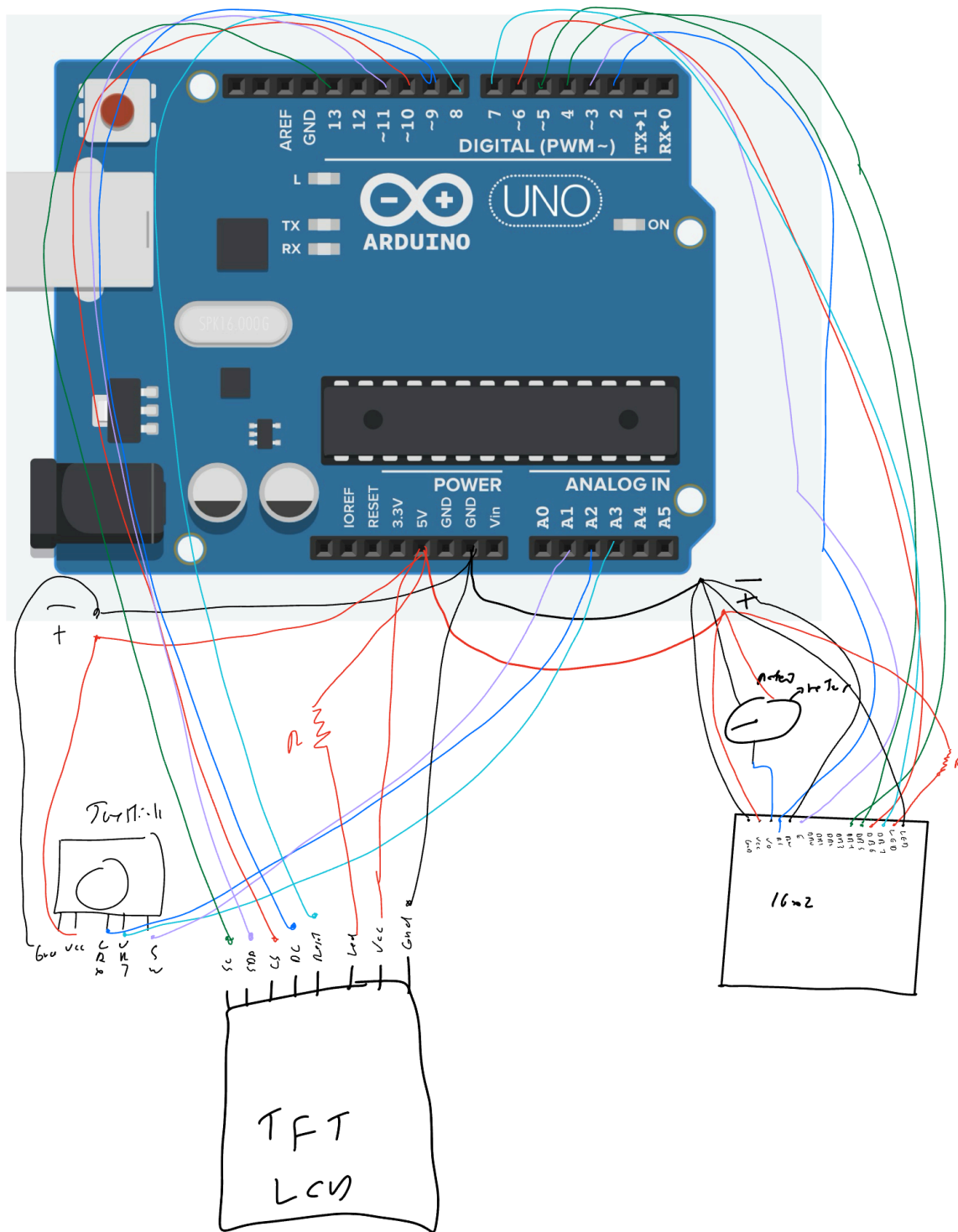
- Arduino UNO R3
- ST7735 1.8" TFT LCD
- 16x2 Character LCD + potentiometer
- Joystick Module
- Breadboard + Jumper Wires
- 220 ohms resistors

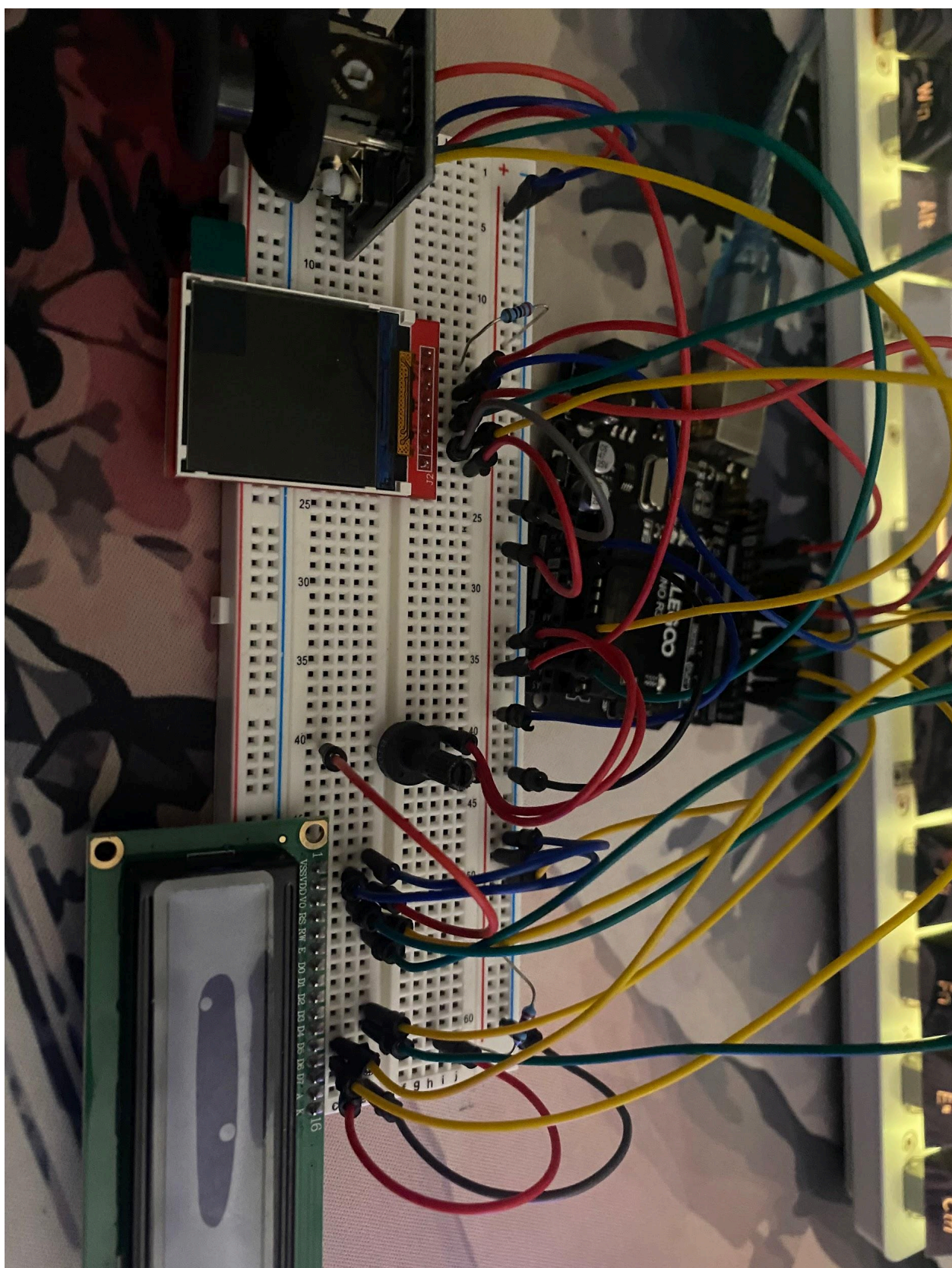
Software Libraries Used

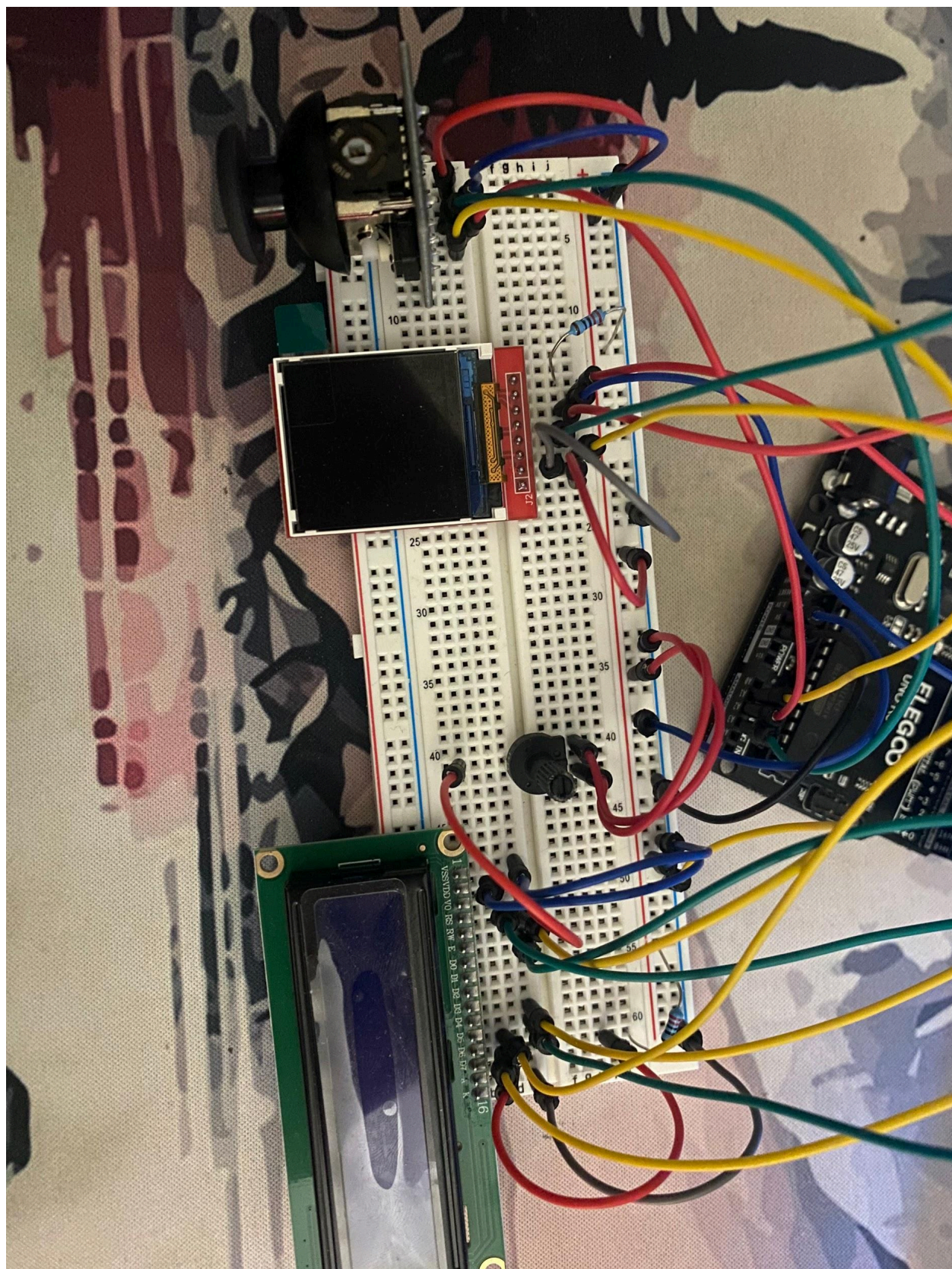
- LCD.h

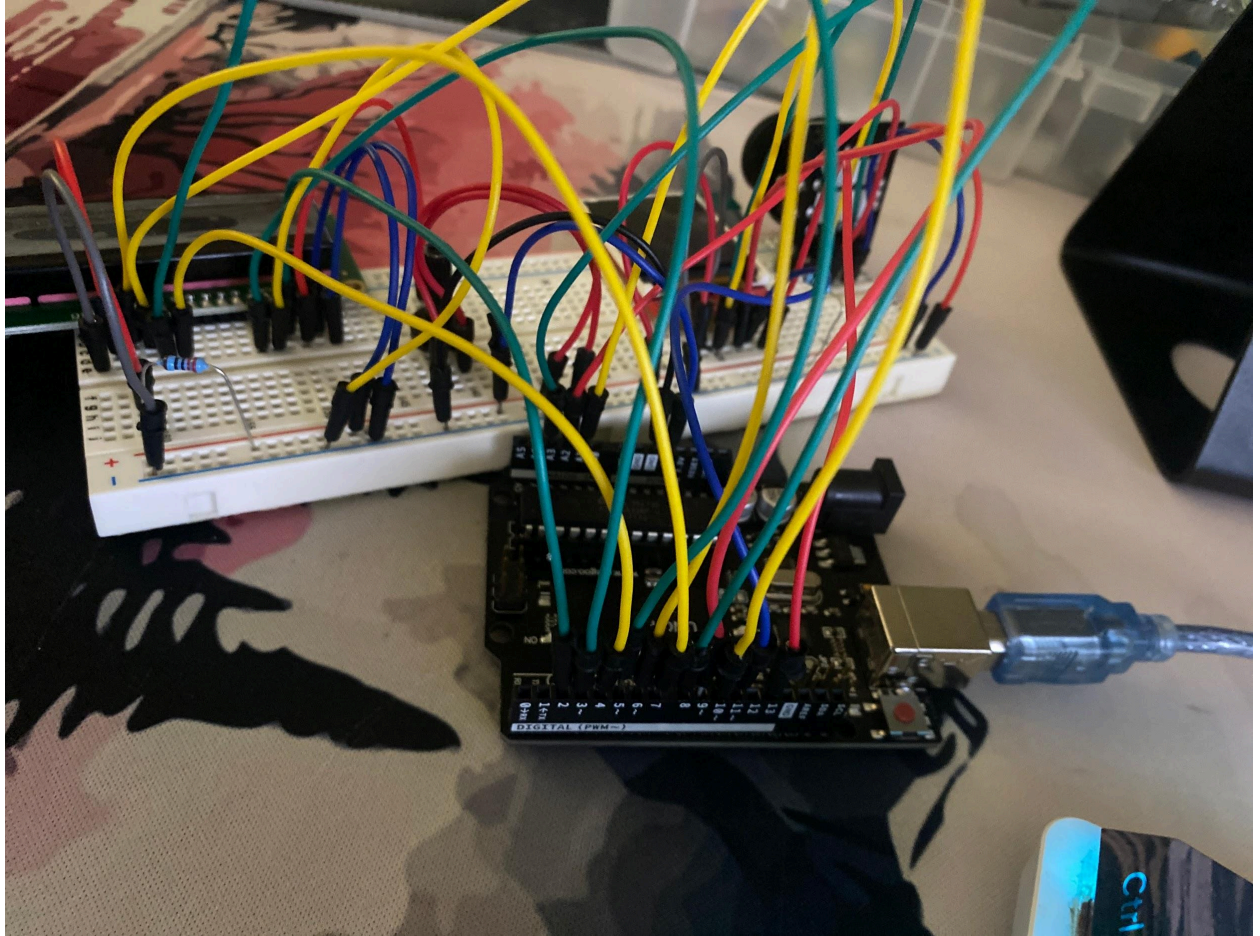
- Provides 16×2 LCD init, clear, write string, and added own custom character code.
- `periph.h`
 - `ADC_init()`. Used for analog joystick readings
- `timerISR.h`
 - Provides the base timer and cooperative task scheduler.
- `phsia006_spiAVR.h`
 - Given SPI driver used to interface the ST7735 LCD plus some own code to finish like full screen and init as well as 18 bit color mode.
- `phsia006_Sprite.h`
 - Contains RGB sprite data for the bird and drawings for pipe

Wiring Diagram





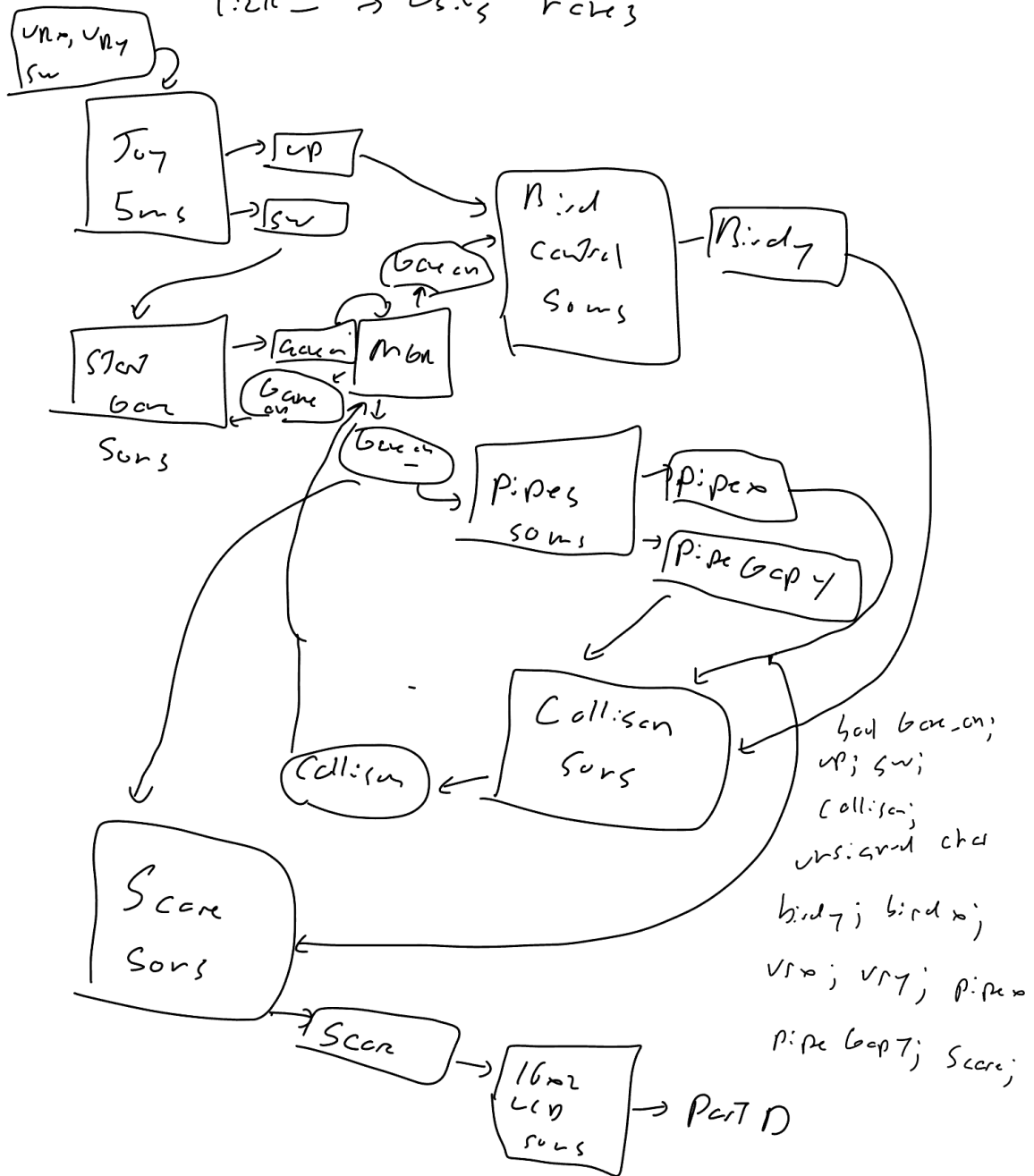




Task Diagram

Task diagram → Since all Task start with

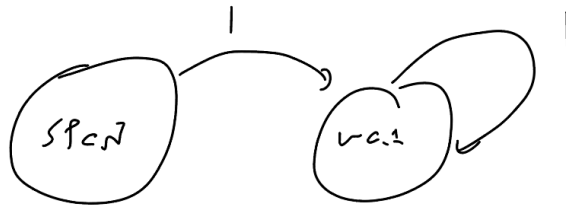
Tick → using roles



SynchSM Diagrams

SynchSM

Task: Top 2 Sems



$vr_7 = \text{DPC_Read}(r)$
 If ($vr_7 < 400$) { $up = 1;$ }
 else if ($vr_7 > 600$) { $down = 1;$ }
 else { $up \wedge down = 0;$ }

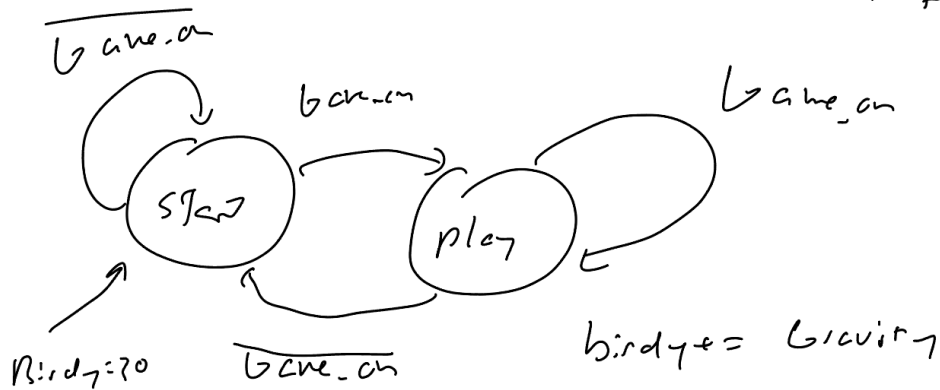
Task: SPC - GCN

Function: `Reset GCN State();`



If ($!GCN_on \ \&\& \ SW$)
 { `Reset GCN State(); GCN_on = 1;` }

Task Bird Control Soms, $unseen \text{ char Gravity} = 1$; Flap = 3

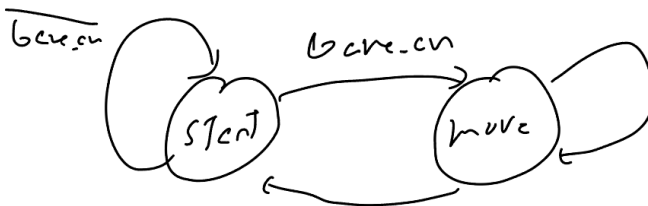


```

birdy += Gravity
if (up) { birdy -= Flapstep; }
if (birdy < min_y) birdy = min_y;
if (birdy > max_y) birdy = max_y;
Draw bird (birdx, birdy)
  
```

Task: Pipes Soms

$unseen \text{ char PipeSpeed} = 1$



$Pipe_x = next.x$ $game.on$

$Pipe Gap = Pipe Gap_{min} + (rand() \cdot (Pipe Gap_{max} - Pipe Gap_{min}))$

$\% (Pipe Gap_{max} - Pipe Gap_{min})$

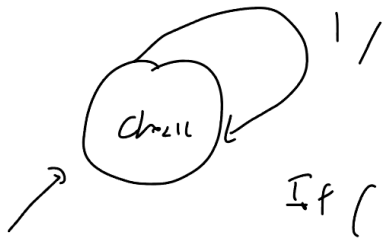
$Pipe Active = 1;$

$game.on / Pipe -= PipeSpeed;$

$If (Pipe_x + PipeWidth > 20) \{$
 $Pipe Active = 0; \}$

$Draw Pipe Pair (Pipe_x,$
 $Pipe Gap);$

Collision Song

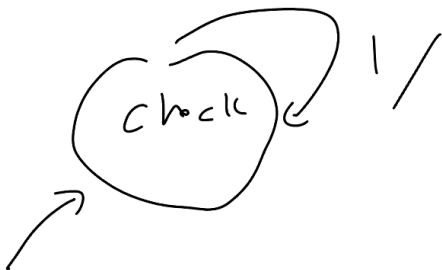


```

uns: and cher birdTop = birdy;
birdLeft = birdy + birdH - 1;
birdRight = birdx;
birdRight = birdx + birdH - 1;
If (Game.ch || pipeActive) {
  If (birdTop ≤ 0 || birdLeft ≥
    ScreenH - 1) { collided = 1; }
  If (birdRight ≥ pipeLeft) && (birdLeft
    ≤ pipeRight) { collided = 1; }
  If (collided) { Game.ch = 0; Game.over = 1; }

```

Score Song



```

If (!Game.ch) { prevPipeActive
  = pipeActive; }
If (prevPipeActive == 1 &&
  pipeActive == 0) { Score++; }

```

LCD16 50ms, unsigned char lcd_node;

